First Step — Evaluation and Test Report

1. Project Information

- **Project:** First Step — an educational fintech prototype for first-time investors in India, built for the Groww GenZ investor assignment.
- **Live application:** https://first-step-groww.vercel.app/
- **Evaluation date:** 10 October 2026, approximately 23:35 IST (18:05 UTC).
- **Source revision:** `7b8523b80d01bedb17c734ec456a67a0eeb61827` (`changes in the website`). The working tree was clean.
- **Testing environment:** Windows PowerShell, Bun, Vitest 4.1.11, Testing Library, and jsdom. The author also tested the site in a real browser at desktop and mobile widths. Playwright, Cypress, and Puppeteer were not installed. No application code was changed during this evaluation.

The commit history includes the session-scoped route gate, consistent end-of-month SIP calculations, and the sticky Flex SIP panel.

2. Executive Summary

Check	Result	Summary
Automated tests	**PASS**	93 of 93 tests passed across 11 test files.
TypeScript type-check	**PASS**	`bunx tsc --noEmit` completed without errors.
Lint on changed and calculation files	**PASS**	No errors or warnings.
Production build	**PASS**	Client, SSR, and Nitro outputs were generated.
SIP calculation verification	**PASS**	Reference values matched the actual calculation function; homepage and Growth Simulator use the same end-of-month contribution timing.
Manual browser evaluation	**PASS, author-reported**	The author reports that all 12 manual scenarios passed on 10 October 2026. These results were not independently observed in this environment.
Deployment	**PASS, with limits**	The public homepage shell loaded and showed the SIP timing fix. Provider-side availability was not independently verified.

3. Automated Testing Results

The following commands were run for this report.

Check	Command	Result	Details
Unit and integration tests	`bun run test`	**PASS**	11 test files; 93 passed, 0 failed, 0 skipped.
TypeScript type-check	`bunx tsc --noEmit`	**PASS**	No errors.
Lint on changed and calculation files	`bunx eslint src/lib/gate.ts src/test/gate.test.ts src/components/landing/MutualFunds.tsx src/routes/flex-sip.tsx src/components/Shell.tsx src/lib/firststep.ts src/lib/learn.ts`	**PASS**	0 errors and 0 warnings.
Production build	`bun run build`	**PASS**	Client build (1,949 modules), SSR build (105 modules), and Nitro outputs generated. Existing advisory warnings were reported for `vite-tsconfig-paths` and Nitro `inlineDynamicImports`.

Automated test coverage

Test file	Tests	Main coverage
`app-routing.test.tsx`	1	Routing
`firststep.test.ts`	29	Readiness outcomes and validation, pause/resume dates, totals, minimum-base rule
`flexsip.test.tsx`	4	Flex plan behavior
`gate.test.ts`	9	Gate store, session-persistence regression tests, memory-router guard
`growthcharts.test.tsx`	4	Growth visuals
`learn.test.ts`	22	SIP calculations, checklist, snapshot, video filtering
`market.test.ts`	10	Market calculations, formatting, and supported universe; no network calls
`myplan.test.tsx`	4	My Plan dashboard
`readiness-gate.test.tsx`	2	Checklist, unlocking, and navigation
`readiness.test.tsx`	5	Readiness assessment
`resources.test.ts`	3	Learning resources
Total	**93**	**All tests passed**

4. SIP Calculation Verification

- **Implementation:** `sipProjection` and `sipValueAtMonth` in `src/lib/learn.ts`.
- **Method:** Future value of a level monthly annuity, using monthly rate `r = annualRatePct / 100 / 12` and `FV = P × (((1 + r)^n − 1) / r)`.
- **Contribution timing:** End-of-month (`"end"`). The homepage sketch, Growth Simulator, and both growth charts use the same convention. The homepage footnote also states that contributions are made at the end of each month.
- **Calculation conventions:** Total contributions equal monthly contribution multiplied by the number of months. Estimated gain equals ending value minus contributions. Zero months or zero contribution returns zero; a zero return rate returns contributions only. Negative inputs are clamped to zero. Cards and charts derive values from the same calculation function rather than hardcoded values.

Independent reference checks

The actual `sipProjection` implementation was imported and checked directly.

Monthly contribution	Duration	Assumed annual return	Total contributed	Expected ending value	Actual ending value	Estimated gain	Match
■1,000	3 years	4%	■36,000	Approximately ■38,182	■38,182	■2,182	**Yes**
■1,000	3 years	8%	■36,000	Approximately ■40,536	■40,536	■4,536	**Yes**
■1,000	3 years	12%	■36,000	Approximately ■43,077	■43,077	■7,077	**Yes**
■2,000	10 years	10%	■2,40,000	Approximately ■4,09,690	■4,09,690	■1,69,690	**Yes**

All figures are illustrative scenarios, not guaranteed or predicted returns. The author also reports that changing inputs updates the figures and charts consistently during manual testing.

5. Manual Evaluation Matrix

Status key: PASS, FAIL, BLOCKED, NOT TESTED.

The following outcomes were reported by the author after real-browser testing on 10 October 2026. They were not independently observed in this environment.

ID	Feature	Test scenario	Expected result	Actual result reported by author	Status
M01	Homepage	Open the deployed URL in a fresh session.	Homepage loads and the intended checklist appears.	Homepage loaded and checklist appeared.	**PASS**
M02	Money Readiness	Enter valid financial information and submit.	The appropriate readiness outcome is displayed.	Correct outcome displayed. Unit tests also cover readiness logic.	**PASS**

ID	Feature	Test scenario	Expected result	Actual result reported by author	Status
M03	Readiness validation	Submit incomplete or invalid values, including zero or negative values where applicable.	Invalid submissions are prevented and validation messages appear.	Invalid submissions were prevented with errors.	**PASS**
M04	Flex SIP	Change base contribution and optional top-up presets.	Totals and summaries update correctly.	Totals updated correctly.	**PASS**
M05	Pause and resume	Save a plan, pause it for 1–3 months, and resume.	Paused state and resume date are correct.	Paused state and resume date were correct.	**PASS**
M06	Growth Simulator	Change contribution amount and duration.	Scenario cards, totals, gains, and visualizations remain consistent.	Values and visuals remained consistent.	**PASS**
M07	My Plan	Save a plan and revisit it.	Saved information appears according to the app's persistence behavior.	Saved plan displayed correctly.	**PASS**
M08	Route refresh	Unlock the app, then refresh each protected route.	Routes remain accessible in the same session.	Routes remained accessible after refresh.	**PASS**
M09	Fresh session	Open a protected route in a new isolated session without unlocking first.	The checklist gate is enforced.	Redirected to the checklist as intended.	**PASS**
M10	Responsive design	Check desktop and mobile layouts.	Content is readable, controls are usable, and there is no horizontal overflow.	Author reported no issues at desktop and mobile widths.	**PASS**
M11	Market-data failure	Exercise an unavailable market-data response.	An honest fallback appears without fabricated data.	Author reported that the fallback appeared. The exact failure-simulation method was not recorded.	**PASS — limited evidence**
M12	Deployment	Open the public URL in a fresh session.	The current deployed app loads.	The public page loaded; the homepage showed the ■2,40,000 contribution and approximately ■4,09,690 illustrative value, along with the end-of-month contribution footnote. The author also confirmed the live app worked.	**PASS — with limits**

Reported manual totals: 12 passed, 0 failed, 0 blocked, 0 not tested.

6. Route-Gating Regression Tests

Automated results use jsdom and a memory router. The real-browser outcomes in the final column are author-reported.

ID	Scenario	Automated result	Real-browser result reported by author
G01	Fresh session	Gate is locked by default; direct `/flex-sip` and nested paths redirect to `/#discover`; `localStorage` cannot unlock it. **PASS**	Gate enforced on fresh access. **PASS**
G02	Unlock	`unlock(3)` fails, `unlock(4)` succeeds, and checklist completion navigates to `/readiness`. **PASS**	Checklist unlocked protected routes. **PASS**
G03	Refresh	Unlock is mirrored to `sessionStorage`; a fresh module load restores it; `lock()` clears it. Two regression tests cover persistence. **PASS**	Refreshed routes remained accessible within the session. **PASS**

ID	Scenario	Automated result	Real-browser result reported by author
G04	Navigation	Navigation from `/risk` to `/flex-sip` remains in the protected area after unlocking. **PASS**	No unexpected redirects. **PASS**
G05	New session	A new tab/session starts locked; unlock is not stored in `localStorage`; server-side rendering guards are present. **PASS**	Fresh session required checklist completion. **PASS**
G06	Back/forward	Back/forward navigation while locked cannot reach gated content. **PASS**	No loss of access after unlocking. **PASS**

7. Market and Mutual Fund Data

Data providers identified in source code

- **Market quotes:** Yahoo Finance chart API for NSE quotes, implemented in `src/lib/market/yahoo.server.ts`. The code includes a 12-second timeout, a five-minute cache, and delayed/closed/stale labels; it does not label quotes as live.
- **Mutual fund NAVs:** `mfapi.in`, implemented in `src/lib/market/mfapi.server.ts`. The code includes a 15-second timeout, a 12-hour cache, and trailing-return calculations based on NAV history through `trailingCagr`.

What was verified

Ten `market.test.ts` unit tests cover calculation and formatting logic, including mover ranking and sorting, volume-shock ratios, NAV CAGR, monthly sampling, Indian number formatting, and handling invalid values such as `NaN` and bad dates. These tests do not make network requests, and the test suite contains no fetch mocks.

Source inspection and author-reported live testing cover loading skeletons, error states, empty categories, source attribution, and demo-versus-live labels. The homepage Market Lab explicitly labels its figures as fictional demo data rather than live prices.

What remains unverified

The external providers' raw live responses and availability were not independently verified. The report therefore does not claim that Yahoo Finance or `mfapi.in` were available at the time of evaluation.

8. Defects and Limitations

Existing lint failure

The full-repository command `bun run lint` fails because of:

- **96 CRLF line-ending errors** in `src/components/PauseControls.tsx`.
- **2 formatting errors** in `src/lib/usePlan.ts`.
- **6 existing React Refresh warnings** in `src/components/ui/`.

The report identifies these files as untouched by the recent changes; their last recorded modification was commit `5f95b4d`. Lint on the recently changed and calculation files passed. The formatting issues were not fixed during this evaluation to avoid unrelated changes.

Other limitations

- No manual scenarios were reported as blocked or untested. However, the market-data failure test did not record exactly how the failure was simulated, and the automated suite has no provider-level failure-injection fixture.
- External availability of Yahoo Finance, `mfapi.in`, YouTube embeds and thumbnails, and Google Fonts was not independently verified.
- Manual browser outcomes are author-reported from 10 October 2026; they were not independently observed by automated browser tooling in this environment.

9. Final Assessment

Result: PASS WITH LIMITATIONS.

The reported functional checks passed: 93 of 93 automated tests, TypeScript type-check, production build, SIP calculation verification, route-gating regression coverage, and all 12 author-reported manual scenarios. The public deployment also showed evidence of the SIP timing fix.

10. Remaining Actions

Resolve the lint failure, if appropriate: run `bunx prettier --write src/components/PauseControls.tsx src/lib/usePlan.ts` (or the project's `bun run format` command), then rerun `bun run lint`. Review the diff before committing. Optionally add a `.gitattributes` rule such as `*.tsx text eol=lf` to help prevent recurring line-ending differences.

Keep supporting evidence: optionally attach dated screenshots or browser notes for route refresh, fresh-session gating, and market-data fallback tests.

Confirm the deployment revision: verify that Vercel is serving commit `7b8523b` or redeploy the intended revision.

Rerun checks after future changes: do not reuse these results as current after additional code changes without rerunning the relevant checks.